PROPRIETARY & CONFIDENTIAL — LWANDILE ENGINEERING PROJECTS & INCLUSIVE ENTREPRENEURIAL ACADEMY

SCAPUSH PRECISION STANDARD

SECTION 30 — CLOSED-LOOP WASTE-TO-VALUE FEEDSTOCK MATRIX

Sovereign Bio-Solar-Thermal Campus | Detailed Engineering Quantification (20,000m² Perimeter Fenced)

Doc Ref: 30_waste_to_value_matrix.html | Status: APPROVED BASELINE (REV 2.0) | Governance: Scapush Precision | Biotic Mass Flow & Circular Economy Integration

Auto-Scrolling Live Narration Engine

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1. Executive Purpose & Biological Mass Balance

LISTEN SECTION

This document establishes the closed-loop waste-to-value mass balance for the **Scaled MVP Baseline (Secured within the 20,000m² perimeter fence, anchored by 4,800 Broilers in Barn 1 and 160 Swine Weaners in Barn 2)**. It quantifies daily organic manure generation, volatile solids (VS) loading rates, biochemical methane potential (BMP), biogas production yields, and high-grade organic bio-fertilizer conversion output to guarantee zero waste discharge and complete circular sovereignty across the securely fenced campus.

BIOTIC CIRCULARITY PRINCIPLE (100% WASTE RE-UTILIZATION)

Organic excreta and washdown slurry from the 4,800 broilers and 160 swine are gravity-fed into the **70m³ mesophilic anaerobic biodigester** located securely within the 20,000m² campus boundary. This biological conversion yields approximately **21.6 m³ of high-purity biogas per day** (fueling the 15 kW CHP engine) and **3.3 m³ of pathogen-free organic liquid digestate per day**, which serves as nutrient-rich bio-fertilizer for campus agriculture.

2. Closed-Loop Waste-to-Value Mass Flow Schematic

LISTEN SECTION

Biotic Mass Flow & Energy Conversion Pathway

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[ Barn 1: 4,800 Broilers ] (Manure Yield: ~240 kg/day inside 20k m² Fence) → Collection Flumes
[ Barn 2: 160 Swine Weaners ] (Slurry Yield: ~640 kg/day) → Gravity Flumes
                                    ↓
[ 70m³ Mesophilic Anaerobic Biodigester Core ] (Target Temp: 38°C)
                                    ↓
├→ Biogas Generation (~21.6 m³/day @ 60% CH4) → 15 kW Biogas CHP Generator (Power + Thermal Heat)
├→ Liquid Digestate (~3,300 L/day) → Multi-Stage Filtration → Certified Bio-Fertilizer for Campus Crops
└→ Recovered Jacket & Exhaust Heat → Barn 1 & 2 Underfloor Hydronic PEX-a Manifolds
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3. Itemized Livestock Waste Generation & Characterization Register

LISTEN SECTION

Livestock Subsystem	Stock Population & Asset Scale	Daily Excreta / Manure Generation Rate	Total Daily Wet Mass (kg / day)	Volatile Solids (VS) Content (%)	Estimated Biogas Yield (m^3 / kg VS)
Barn 1 (Poultry)	4,800 Broilers (Average live weight 1.8 kg)	0.05 kg / bird / day	**240.0 kg / day**	78.0 % VS	0.35 m^3 / kg VS
Barn 2 (Swine)	160 Weaner Pigs (Average live weight 30 kg)	4.00 kg / animal / day	**640.0 kg / day**	82.0 % VS	0.42 m^3 / kg VS
		TOTAL DAILY RAW BIOMASS INPUT:	**880.0 kg / day Wet Organic Mass**		

4. Biodigester Mass Balance, Biogas Yield & Bio-Fertilizer Output

LISTEN SECTION

Biodigester Input & Biogas Production

- **Total Daily Wet Feedstock:** 880 kg manure + 900 L dilution water = **1,780 L/day slurry**.
- **Hydraulic Retention Time (HRT):** 32 days within the 70m^3 tank volume.
- **Total Daily Biogas Production:** ~21.6 m^3 / day (methane concentration 58% – 62%).
- **CHP Electrical Equivalent:** Fueling the 15 kW CHP engine for 1.4 hours of baseload generation daily (~21 kWh electrical).

Bio-Fertilizer & Nutrient Recovery

- **Daily Liquid Digestate Output:** ~3,300 L/day (including recycled greywater inputs).
- **Pathogen Reduction:** 99.9% pathogen kill via anaerobic mesophilic digestion (38°C) and secondary settling.
- **Nutrient Profile:** High concentration of bio-available nitrogen (N), phosphorus (P), and potassium (K).
- **Environmental Impact:** Zero chemical fertilizer runoff; 100% closed-loop nutrient cycling across the 20,000m^2 campus.

5. Waste-to-Value Architecture Frozen Lock-In Register

LISTEN SECTION

Ref Code	Locked Waste-to-Value Parameter	Engineering Specification & Governance Basis
WTV-001	Poultry Manure Input	240 kg/day raw organic manure from 4,800 broilers in Barn 1.
WTV-002	Swine Manure Input	640 kg/day wet slurry and manure from 160 swine weaners in Barn 2.
WTV-003	Total Daily Wet Biomass	880 kg/day combined livestock excreta feeding the 70m^3 biodigester.
WTV-004	Biogas Generation Yield	~21.6 m^3/day high-purity biogas fueling co-generation thermal loops.
WTV-005	Organic Bio-Fertilizer Output	~3,300 L/day pathogen-free liquid digestate returned to campus agriculture.
WTV-006-F	20,000m^2 Fenced Circular Biosecurity	Complete containment of all organic biomass feedstocks within the secure 20,000m^2 perimeter boundary.

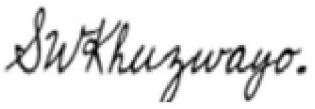
6. Approval & Authorization

LISTEN SECTION

DOCUMENT STATUS

STATUS: APPROVED BASELINE (REV 2.0) — 20,000M² FENCE INFUSED & LOCKED FOR WASTE-TO-VALUE AUDIT

PROJECT STEWARD & TECHNICAL LEAD



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Consortium Leadership

Sipho Wiseman Khuzwayo — Lead System Architect (Lwandile Engineering)
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Execution Entities & Standard

Lwandile Engineering Projects (Pty) Ltd
Inclusive Entrepreneurial Academy
Scapush Precision Execution Standard

Operational Footprint

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